1. What HCI / Interaction Design Really Is

Human–Computer Interaction (HCI) is about **how people and computers communicate** — not through code, but through **interfaces**: screens, buttons, gestures, sounds, feedback, layouts, etc.

Key idea:

> We don't design computers.
> **We design interactions.**

Core definitions (VERY IMPORTANT)

Sharp, Rogers & Preece (2019):

> Designing interactive products to support how people communicate and interact in their everyday and working lives.

Winograd (1997):

> Design of spaces for human communication and interaction.

👉 Exam trick:
Always include **"everyday + working lives"** or **"spaces for communication"**.

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2. Related Design Terms (MEMORIZE SHORT DEFINITIONS)

Term	Meaning
User-Centered Design (UCD)	Focus on users, tasks, and usability
Human-Centered Design (HCD)	Broader: cognitive, physical, emotional needs
People-Centered Design	Informal HCD, real everyday people
Customer Experience (CX)	Entire journey with a company (ads → app → support)

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3. Good vs Bad Design (HCI Mindset)

HCI constantly asks:
“Why is this confusing?” or **“Why does this feel natural?”**

Good Design Example – TiVo Remote

Why it works:

- Peanut shape fits the hand
- Buttons grouped logically
- Color + shape cues
- Becomes usable **without looking**

Bad Design Patterns (What examiners LOVE)

- **Elevator buttons**: labels look pressable → poor visibility/signifiers
- **Vending machine**: breaks user mental model (money first)
- **Road sign ambiguity**: conflicting symbols
- **Identical buttons**: volume vs channel triangles
- **Decorative calendars**: aesthetics destroy function

👉 Rule:

> Decoration must **never destroy the primary task**.

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4. Cognition in HCI (VERY IMPORTANT)

Cognition = mental processes:

- Thinking
- Perception
- Attention
- Memory
- Learning
- Reasoning

Why cognition matters:

Poor design increases **cognitive load**, causing:

- Errors
- Stress
- Accidents (e.g., aviation, driving, medical systems)

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5. Attention

- Humans have **limited attention**
- Multitasking ≠ parallel thinking
- Interfaces must **guide attention**, not demand it

Design implications:

- Highlight important info
- Reduce clutter
- Avoid forcing users to remember things while acting

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6. Perception

People don’t see the world objectively — perception depends on:

- Contrast
- Grouping
- Expectation
- Context

Key ideas:

- Color alone is unreliable (color blindness)
- Shape, position, labels must reinforce meaning

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7. Memory (EXAM GOLD)

Three-Stage Model

1. **Sensory memory** (milliseconds)
2. **Short-Term / Working memory** (limited capacity)
3. **Long-Term memory** (knowledge, habits)

Recognition vs Recall

- **Recognition**: easier (menus, icons)

- **Recall**: harder (typing commands)

👉 Design rule:

> Always prefer **recognition over recall**

Miller's 7±2 (COMMON MISUSE)

- Not a menu design rule
- Chunking matters more than number

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8. Mental Models

Mental model = user's internal understanding of how a system works.

Problems occur when:

- Designer model ≠ User model

Good design:

- Matches real-world expectations
- Uses familiar metaphors

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9. Gulfs of Interaction (Norman)

- **Gulf of Execution**: Can user figure out what to do?
- **Gulf of Evaluation**: Can user understand what happened?

Good interfaces:

- Reduce both gulfs using feedback, visibility, consistency

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10. Distributed, External & Embodied Cognition

- **Distributed cognition**: thinking spread across people + tools
- **External cognition**: notes, calendars, reminders help memory
- **Embodied cognition**: body + environment affect thinking

Example:
Post-it notes, whiteboards, gestures, physical controls

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11. Design Process (VERY TESTED)

Stages

1. Conceptual design (what is it?)
2. Physical design (how it looks/works)
3. Implementation
4. Evaluation (always ongoing)

Technology Myopia (IMPORTANT TERM)

Focusing on tech instead of:

- Social context
- Human habits
- Emotions
- Workarounds

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12. Prototyping

Why prototype?

- Test ideas early
- Save development cost
- Communicate with stakeholders
- Find usability issues early

Sketch → Wireframe → Prototype

- **Sketch**: fast, disposable ideas
- **Wireframe**: structure & flow
- **Prototype**: interaction simulation

Fidelity

- **Low-fidelity**: paper, sketches

- **High-fidelity**: interactive, realistic

Trade-off:

- Lo-fi = fast, flexible
- Hi-fi = realistic, costly, risk of sunk cost

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13. Evaluation & Research

Why evaluate?

- Designers are not users
- Assumptions fail

Key Methods

- Ethnography (observe in real context)
- Interviews
- Surveys
- Focus groups
- Card sorting
- Walkthroughs
- Usability testing
- Heuristic evaluation

Triangulation

Using **multiple methods** to increase validity.

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14. Usability Goals (MEMORIZE LIST)

1. Effective
2. Efficient
3. Safe
4. Good utility
5. Easy to learn
6. Easy to remember
7. Enjoyable

Usability vs UX

- Usability = measurable
- UX = emotional, subjective

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15. UX & Emotional Design

Hassenzahl's Model

- **Pragmatic**: usefulness, efficiency
- **Hedonic**: pleasure, stimulation, meaning

Norman's 3 Levels

1. **Visceral** – look & feel
2. **Behavioral** – usability in action
3. **Reflective** – meaning, identity, memory

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16. Emotional Interaction

- Emotions affect cognition
- Negative emotions → narrow thinking
- Positive emotions → creativity

Design implications:

- Error messages matter
- Tone matters
- Anxiety-heavy systems (ATM, banking) need clarity

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17. Design Principles (HUGE)

Principle	Meaning
Visibility	Make actions obvious
Feedback	Tell user what happened

Constraints	Prevent wrong actions
Mapping	Controls match effects
Consistency	Same actions behave same
Affordance	Suggest how to use

👉 In GUIs: **perceived affordances** matter.

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18. Accessibility & Inclusiveness

Accessibility

Usable by people with disabilities.

Inclusiveness

Usable by the widest range of people.

Disability Types

- Sensory
- Physical
- Cognitive

Duration

- Permanent
- Temporary
- Situational

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19. Culture & Context

- Date formats differ
- Symbols differ
- Language, color, layout meanings vary

Design must consider **local + global users**.

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20. Mobile HCI

Carry Principle

Mobile = user moves, not device.

Characteristics:

- Interruptible
- Contextual
- Personal
- Always available

Apple HIG

- Clarity
- Deference
- Depth

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21. Advanced HCI

Ubiquitous Computing

Computers embedded in environment.

Interaction styles:

- Implicit input
- Context awareness
- Ambient feedback

AR vs VR

- VR replaces reality
- AR augments reality